

WhatsNext Vision Motors Salesforce Project Documentation

Prepared for Project Review

Salesforce Platform Developer Project

Data Model Automation Apex Batch Processing Testing

WhatsNext Vision Motors

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This report presents the WhatsNext Vision Motors Salesforce implementation in a concise review format. Each numbered figure identifies the screen or evidence to be inserted from the Salesforce Developer Edition environment.

1 Project Overview

WhatsNext Vision Motors is a Salesforce-based automotive management solution designed to bring vehicle, dealer, customer, order, test-drive, and service information into one platform. The project focuses on making common vehicle-sales activities more organized and reducing repetitive manual work.

The implementation combines Salesforce custom objects and relationships with Record-Triggered Flows, Apex Triggers, Batch Apex, and Scheduled Apex. Together, these features automate important parts of the ordering and customer-service process.

A major business requirement is vehicle stock validation. Before an order is processed, the system checks whether the selected vehicle has available stock. If the stock is unavailable, the order is prevented from being placed. When an order is confirmed, the vehicle stock is reduced accordingly.

The project also includes dealer assignment based on customer location information and an automated reminder for scheduled test drives. Pending orders can be revisited through Batch Apex after new vehicle stock becomes available.

Main Business Outcomes

- Centralize vehicle, dealer, customer, order, test-drive, and service information.
- Prevent vehicle orders when the selected vehicle has no available stock.
- Associate orders with the relevant dealer using customer location information.
- Send an automated reminder before a scheduled test drive.
- Update vehicle stock when a confirmed order is processed.
- Process pending orders in bulk after stock becomes available.
- Run pending-order processing automatically through a scheduled Apex job.

Technology Used

Area	Salesforce Feature
Data Management	Custom Objects, Custom Fields, Lookup Relationships
User Interface	Custom Tabs, Lightning App
Automation	Record-Triggered Flow, Scheduled Path
Development	Apex, Trigger, Trigger Handler
Bulk Processing	Batch Apex
Scheduling	Scheduled Apex
Communication	Automated Email

2 Objectives and Implementation Scope

Project Objectives

- Create a structured Salesforce data model for automotive operations.
- Maintain vehicle and stock information in a centralized system.
- Manage customer details and their vehicle-related activities.
- Track vehicle orders and their status.
- Prevent orders for vehicles that are out of stock.
- Automate dealer assignment using customer location information.
- Manage test-drive bookings and send automated reminders.

- Reduce vehicle stock when confirmed orders are processed.
- Process pending orders automatically when stock becomes available.
- Demonstrate practical use of Salesforce configuration and Apex development.

Implementation Scope

Area	Included Implementation	Review Evidence
Data Model	Six custom objects with fields and lookup relationships	Object Manager / Schema Builder
Application	Custom tabs and Lightning App	Lightning App screen
Dealer Assignment	Record-Triggered Flow	Flow Builder
Test Drive Reminder	Scheduled Path and Send Email action	Flow Builder / Email evidence
Order Validation	Apex Trigger + Handler	Developer Console
Stock Update	Apex Trigger + Handler	Vehicle record before/after
Bulk Processing	Batch Apex	Apex Jobs / order records
Scheduling	Scheduled Apex	Scheduled Jobs
Testing	Business scenarios and results	Screenshots

3 Requirement Analysis and Data Model

3.1 Requirement Analysis

The system was designed around the operational needs of an automotive company. The business needs a simple way to maintain vehicle inventory, dealer information, customer details, orders, test-drive bookings, and service requests.

Each major business entity is represented by a custom Salesforce object. Lookup relationships connect these records so that users can move from a customer to an order, from an order to a vehicle, and from a vehicle-related activity to the relevant customer and vehicle.

3.2 Custom Objects

Custom Object	Key Information	Business Purpose
Vehicle__c	Vehicle name, model, stock, price, status	Stores vehicle details and availability
Vehicle_Dealer__c	Dealer name, location, code, phone, email	Stores authorized dealer information
Vehicle_Customer__c	Customer name, email, phone, address, preferred type	Stores customer information
Vehicle_Order__c	Customer, vehicle, order date, status	Tracks vehicle purchases
Vehicle_Test_Drive__c	Customer, vehicle, date, status	Tracks test-drive bookings
Vehicle_Service_Request__c	Customer, vehicle, service date, issue, status	Tracks vehicle service requests

3.3 Object Relationships

Child Object	Relationship	Related Object	Purpose
Vehicle Order	Lookup	Vehicle Customer	Connects an order to the purchasing customer
Vehicle Order	Lookup	Vehicle	Identifies the vehicle being ordered
Vehicle Test Drive	Lookup	Vehicle Customer	Connects a test drive to its customer
Vehicle Test Drive	Lookup	Vehicle	Identifies the vehicle for the test drive
Vehicle Service Request	Lookup	Vehicle Customer	Connects a service request to its customer
Vehicle Service Request	Lookup	Vehicle	Identifies the vehicle requiring service
Vehicle	Lookup	Vehicle Dealer	Associates vehicle information with a dealer

3.4 Key Fields by Object

Object	Important Fields	Purpose
Vehicle__c	Vehicle Name, Vehicle Model, Stock__c, Price__c, Dealer__c, Status__c	Vehicle inventory and availability
Vehicle_Dealer__c	Dealer Name, Dealer Location, Dealer Code, Phone No, Email	Dealer identification and contact
Vehicle_Customer__c	Customer Name, Email, Phone No, Address, Preferred Vehicle Type	Customer information and preference
Vehicle_Order__c	Customer, Vehicle, Order Date, Status, Vehicle Order No	Purchase/order tracking
Vehicle_Test_Drive__c	Customer, Vehicle, Test Drive Date, Status, Test Drive No	Test-drive tracking
Vehicle_Service_Request__c	Customer, Vehicle, Service Date, Issue, Status, Service Request No	Service request tracking

SETUP > OBJECT MANAGER
Vehicle

Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
List View Button Layout

Fields & Relationships
9 Items, Sorted by Field Label

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Dealer	Dealer__c	Lookup(Vehicle Dealer)		✓
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Price	Price__c	Currency(18, 0)		
Status	Status__c	Picklist		
Stock Quantity	Stock__c	Number(18, 0)		
Vehicle Model	Vehicle_Model__c	Picklist		

Figure 1 Vehicle__c Fields & Relationships

The Vehicle object contains the core inventory fields used by the project. The screenshot shows the Dealer lookup, Price, Status, Stock Quantity (API name Stock__c), and Vehicle Model fields configured in Salesforce.

SETUP > OBJECT MANAGER
Vehicle Customer

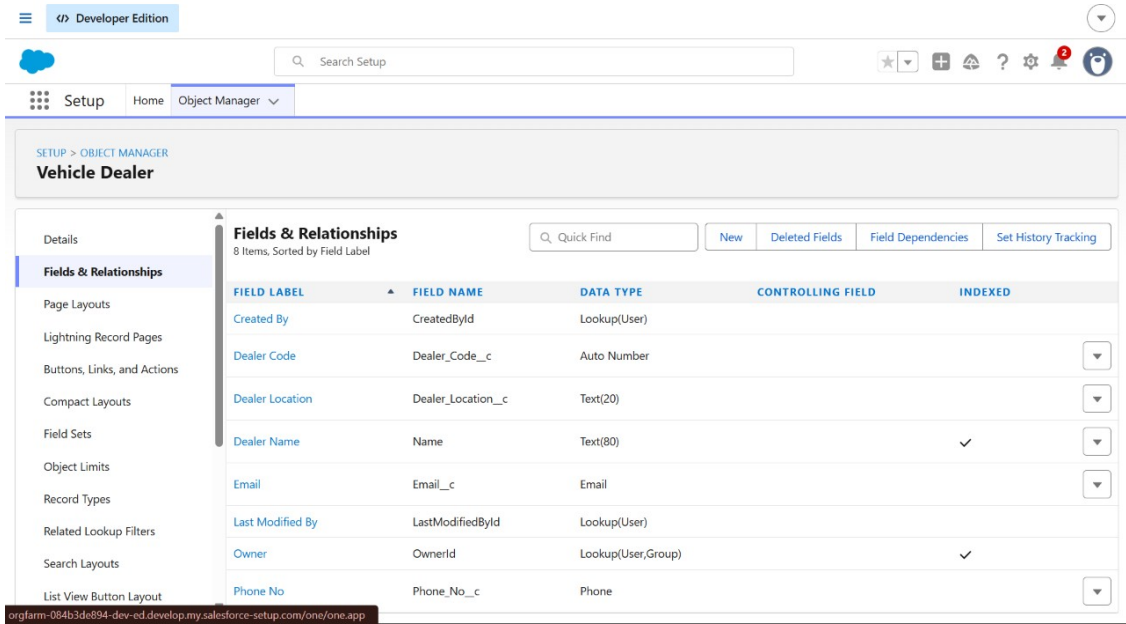
Details
Fields & Relationships
Page Layouts
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Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
List View Button Layout

Fields & Relationships
8 Items, Sorted by Field Label

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Address	Address__c	Text(100)		
Created By	CreatedById	Lookup(User)		
Customer Name	Name	Text(80)		✓
Email	Email__c	Email		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Phone No	Phone_No__c	Phone		
Preferred Vehicle Type	Preferred_Vehicle_Type__c	Picklist		

Figure 2 Vehicle Customer Fields & Relationships

The Vehicle Customer object stores customer contact and preference information. The configured fields include Address, Customer Name, Email, Phone No, and Preferred Vehicle Type.



SETUP > OBJECT MANAGER

Vehicle Dealer

Details

Fields & Relationships

8 Items, Sorted by Field Label

Q Quick Find

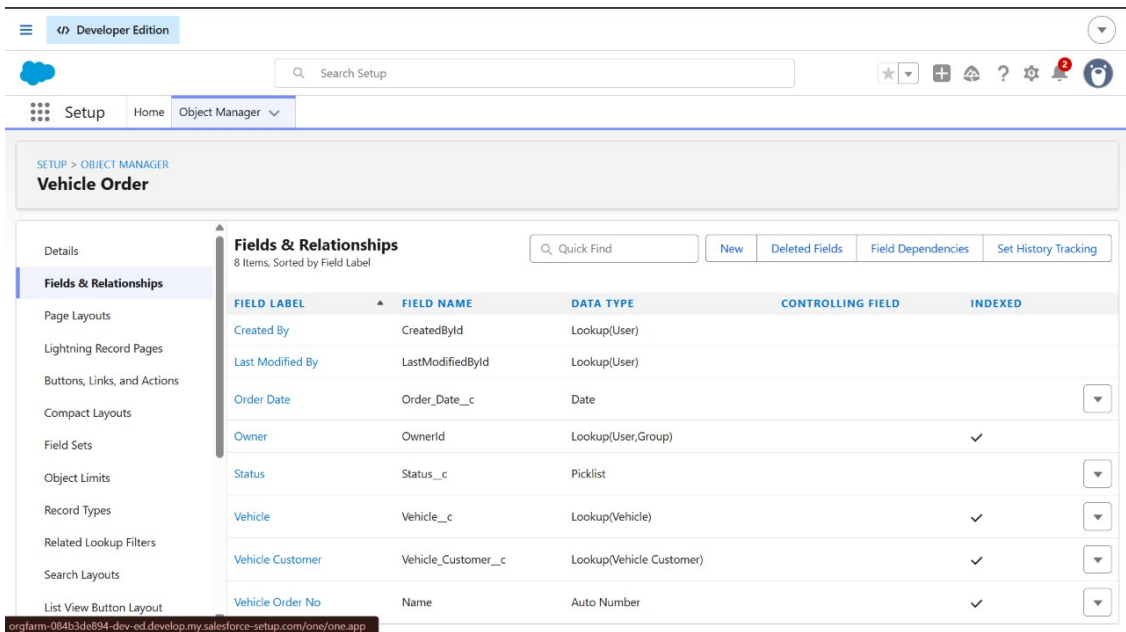
New Deleted Fields Field Dependencies Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Dealer Code	Dealer_Code__c	Auto Number		
Dealer Location	Dealer_Location__c	Text(20)		
Dealer Name	Name	Text(80)		✓
Email	Email__c	Email		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Phone No	Phone_No__c	Phone		

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Figure 3 Vehicle Dealer Fields & Relationships

The Vehicle Dealer object stores dealer identification and contact information. Dealer Code is maintained as an Auto Number, while location, name, email, and phone are stored for dealer management.



SETUP > OBJECT MANAGER

Vehicle Order

Details

Fields & Relationships

8 Items, Sorted by Field Label

Q Quick Find

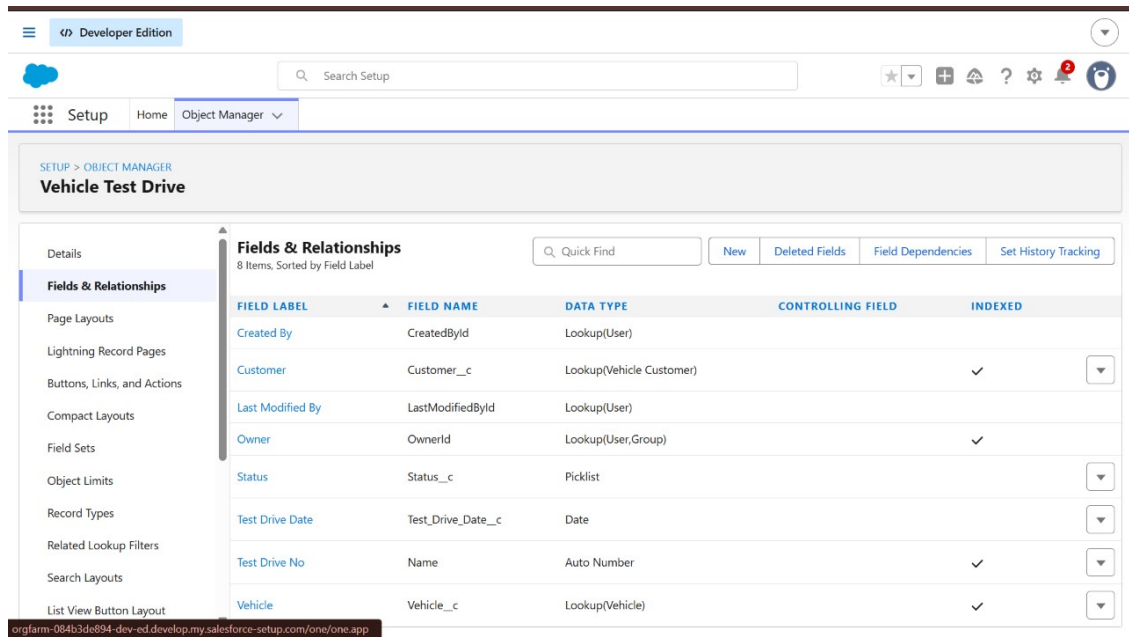
New Deleted Fields Field Dependencies Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Last Modified By	LastModifiedById	Lookup(User)		
Order Date	Order_Date__c	Date		
Owner	OwnerId	Lookup(User,Group)		✓
Status	Status__c	Picklist		
Vehicle	Vehicle__c	Lookup(Vehicle)		✓
Vehicle Customer	Vehicle_Customer__c	Lookup(Vehicle Customer)		✓
Vehicle Order No	Name	Auto Number		✓

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Figure 4 Vehicle Order Fields & Relationships

The Vehicle Order object connects the customer and vehicle and records the order date and status. The Vehicle Order Number is generated as an Auto Number.



SETUP > OBJECT MANAGER

Vehicle Test Drive

Details

Fields & Relationships
8 Items, Sorted by Field Label

Quick Find

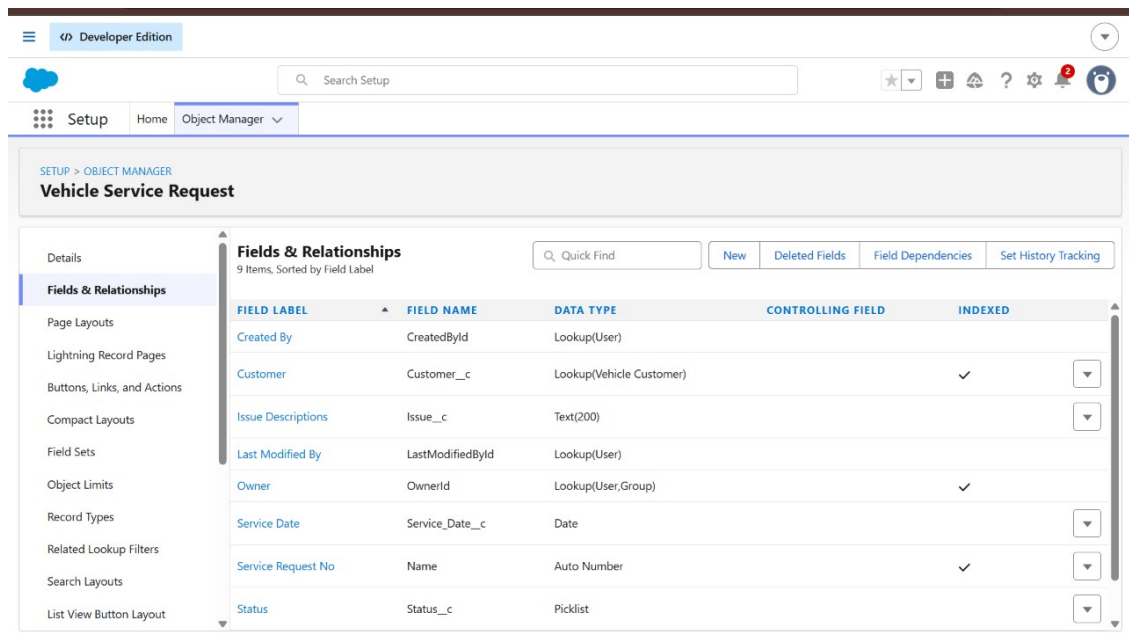
New Deleted Fields Field Dependencies Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Customer	Customer__c	Lookup(Vehicle Customer)		✓
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Status	Status__c	Picklist		
Test Drive Date	Test_Drive_Date__c	Date		
Test Drive No	Name	Auto Number		✓
Vehicle	Vehicle__c	Lookup(Vehicle)		✓

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Figure 5 Vehicle Test Drive Fields & Relationships

The Vehicle Test Drive object connects the customer and vehicle and stores the scheduled date and test-drive status. The Test Drive Number is generated automatically.



SETUP > OBJECT MANAGER

Vehicle Service Request

Details

Fields & Relationships
9 Items, Sorted by Field Label

Quick Find

New Deleted Fields Field Dependencies Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Customer	Customer__c	Lookup(Vehicle Customer)		✓
Issue Descriptions	Issue__c	Text(200)		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Service Date	Service_Date__c	Date		
Service Request No	Name	Auto Number		✓
Status	Status__c	Picklist		

Figure 6 Vehicle Service Request Fields & Relationships

The Vehicle Service Request object stores the customer, service date, issue description, and service status. The service request number is maintained as an Auto Number.

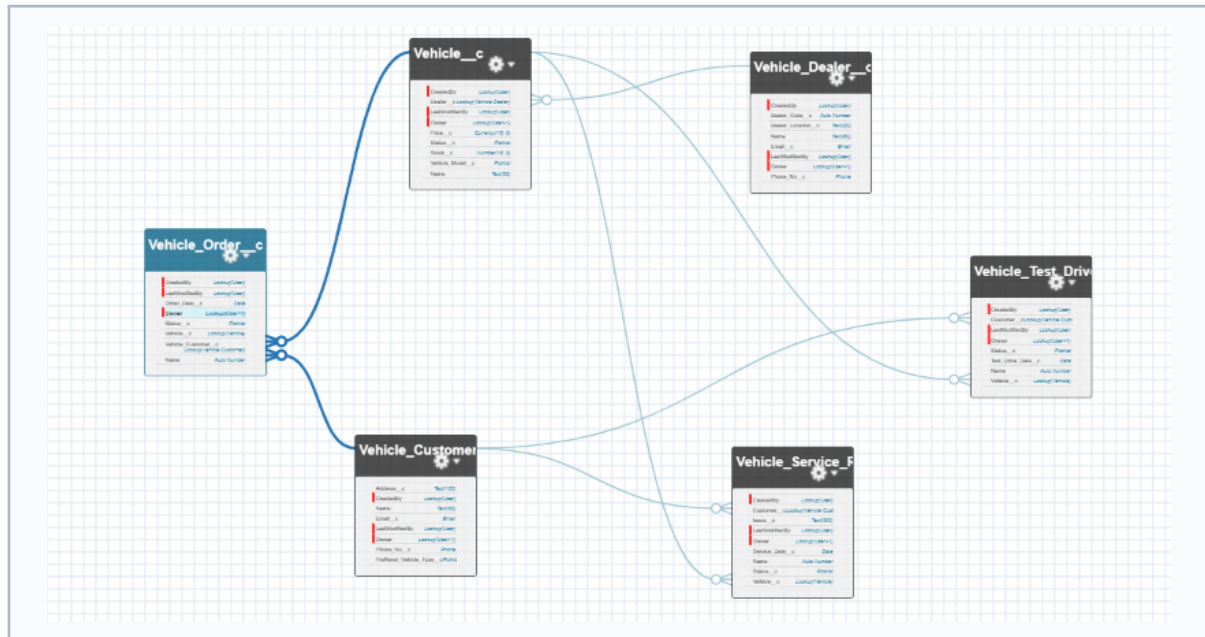


Figure 7 Schema Builder Data Model

4 Salesforce Configuration and Application Setup

4.1 Custom Tabs

After the six custom objects were created, custom tabs were configured so that the records could be accessed directly from the Salesforce application. This provides a simple navigation experience for users working with the project.

- Vehicle
- Vehicle Dealer
- Vehicle Customer
- Vehicle Order
- Vehicle Test Drive
- Vehicle Service Request

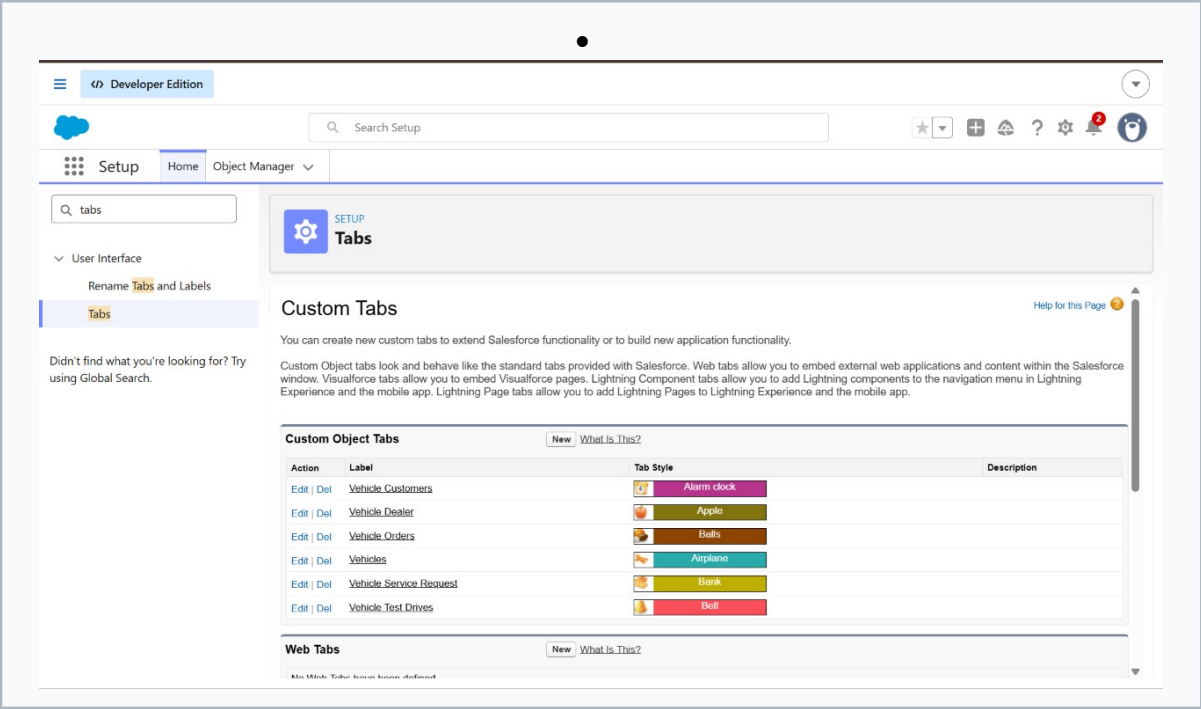
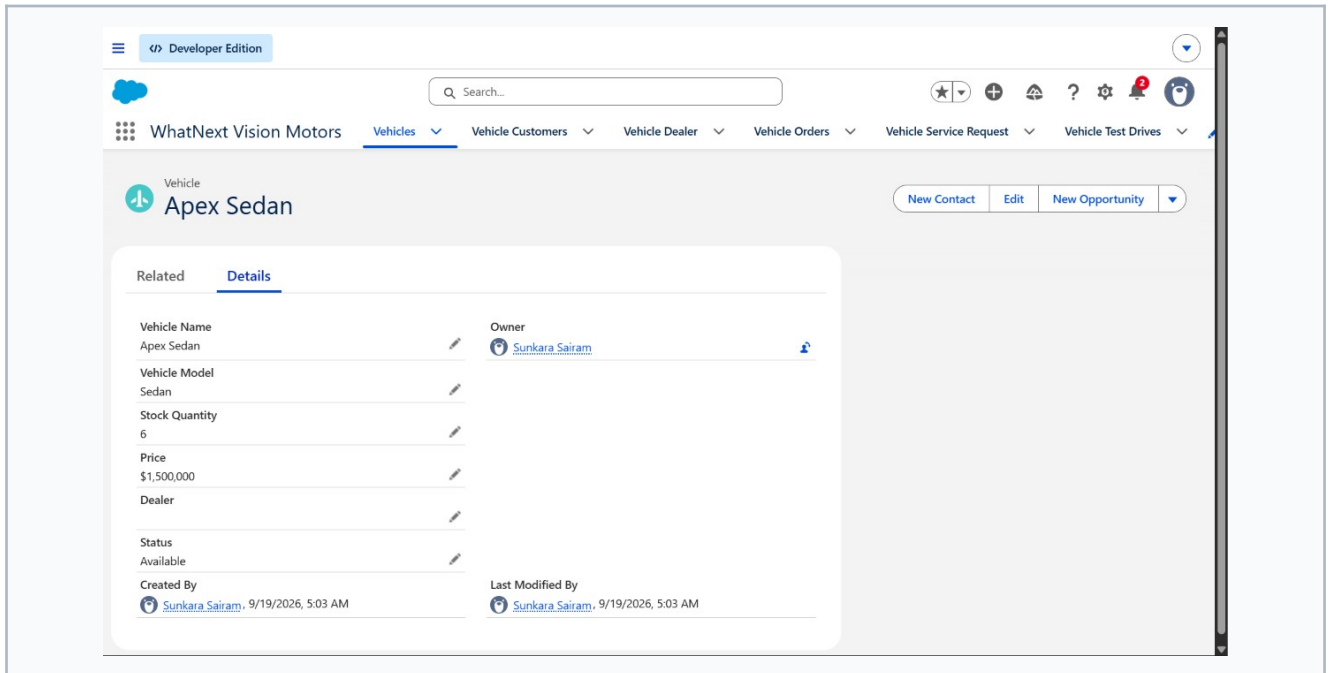


Figure 8 Custom Tabs

4.2 Lightning App

A dedicated Lightning App was created to bring the project objects together in one workspace. The application provides a central navigation area for vehicles, dealers, customers, orders, test drives, and service requests.

Figure 9 WhatsNext Vision Motors Lightning App



5 Data Management and Business Rules

5.1 Vehicle Stock Validation

Vehicle stock is one of the main business rules in the project. When an order is created or updated, the Apex Trigger Handler checks the selected vehicle and reads its Stock__c value.

If the available stock is zero or below, the order receives an error and the transaction is stopped. This ensures that an unavailable vehicle cannot be ordered.

```

Order
↓
Find Vehicle
↓
Check Stock__c
↓
Stock__c <= 0 ?
↓
Yes → Block Order
↓
Display: This vehicle is out of stock. Order cannot be placed.
    
```

5.2 Order Status

Vehicle orders use Pending, Confirmed, Delivered, and Canceled as the available status values. Pending is used for orders that still require stock-based processing.

5.3 Stock Update

When a vehicle order is confirmed, the Apex logic reduces the related vehicle stock by one. This keeps the inventory value aligned with confirmed purchases.

```

Order Status = Confirmed
↓
Vehicle Stock__c = Vehicle Stock__c - 1

```

5.4 Pending Order Processing

When a vehicle has no stock, the order can remain Pending. After inventory is replenished, the Batch Apex process checks the pending order and can confirm it when stock is available.

6 Salesforce Flows and Process Automation

6.1 Auto Assign Dealer Flow

The Auto Assign Dealer Flow is a Record-Triggered Flow on Vehicle_Order__c. It starts when a new order is created with Pending status. The Flow retrieves the related customer and uses the customer's address together with dealer location information to identify the relevant dealer record.

Flow Configuration

Setting	Configured Value
Object	Vehicle_Order__c
Trigger	Record is Created
Entry Condition	Status__c Equals Pending
Get Records 1	Get Customer Information
Get Records 2	Get Nearest Dealer
Purpose	Associate the relevant dealer with the order

Flow Sequence

```

Vehicle Order Created
↓
Status = Pending
↓
Get Customer Information
↓
Get Nearest Dealer
↓
Update / Associate Dealer with Order

```

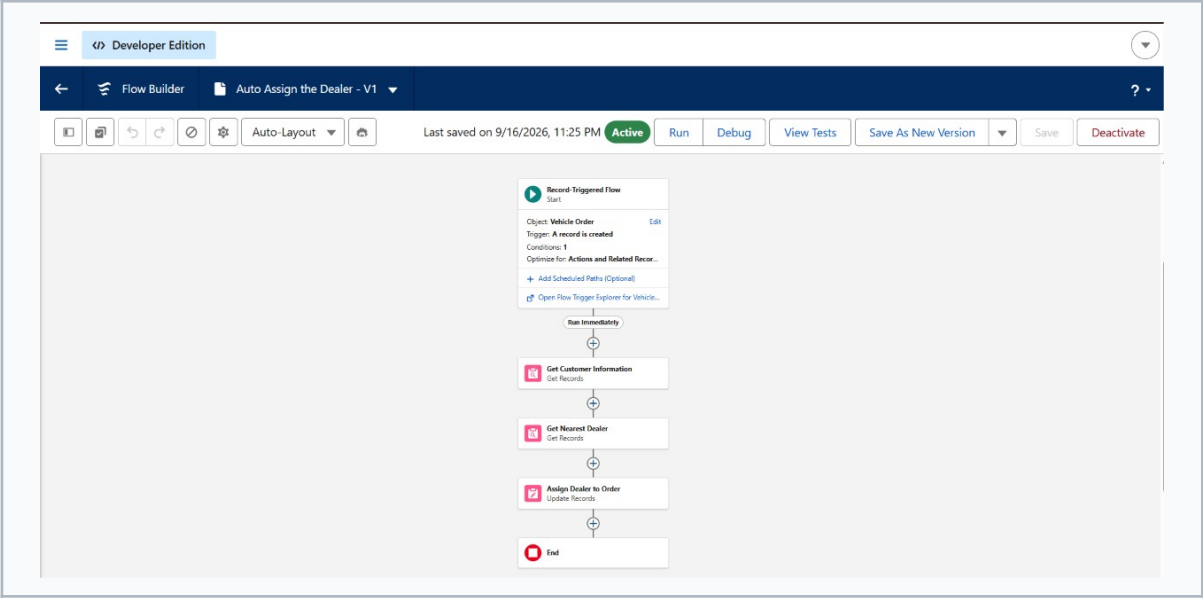


Figure 10 Auto Assign Dealer Flow

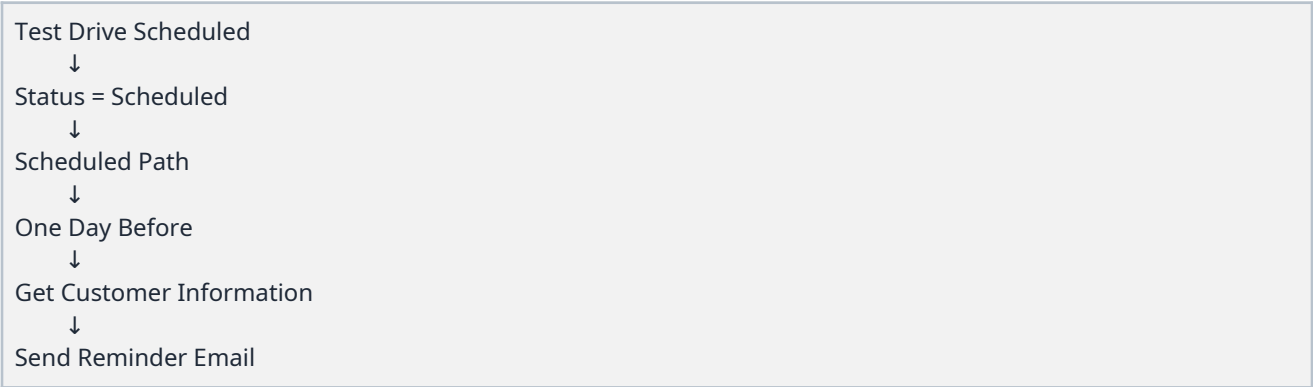
6.2 Test Drive Reminder Flow

The Test Drive Reminder Flow is a Record-Triggered Flow on Vehicle_Test_Drive__c. It runs when a test-drive record is created or updated and the status is Scheduled.

Flow Configuration

Setting	Configured Value
Object	Vehicle_Test_Drive__c
Trigger	Record is Created or Updated
Entry Condition	Status__c Equals Scheduled
Scheduled Path	One day before Test_Drive_Date__c
Action	Send Email
Email Subject	Reminder: Your Test Drive is Tomorrow!

Flow Sequence



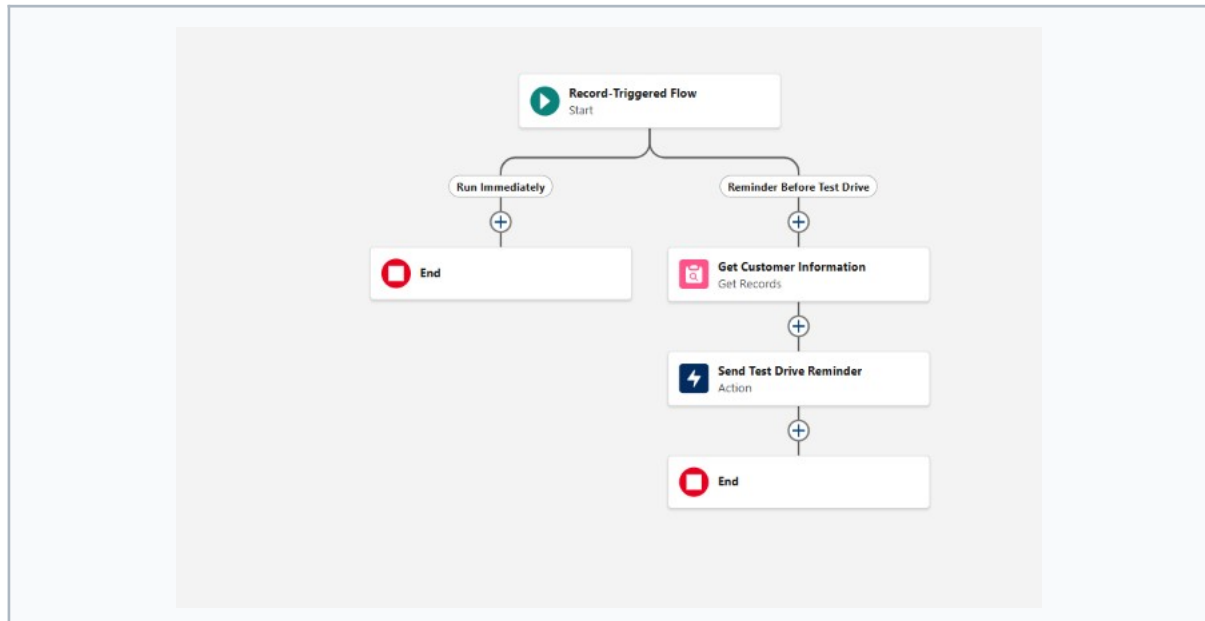


Figure 11 Test Drive Reminder Flow

Figure 12 Automated Test Drive Reminder Email

Evidence screenshot: email-received view to be inserted after successful delivery.

7 Apex Trigger and Trigger Handler

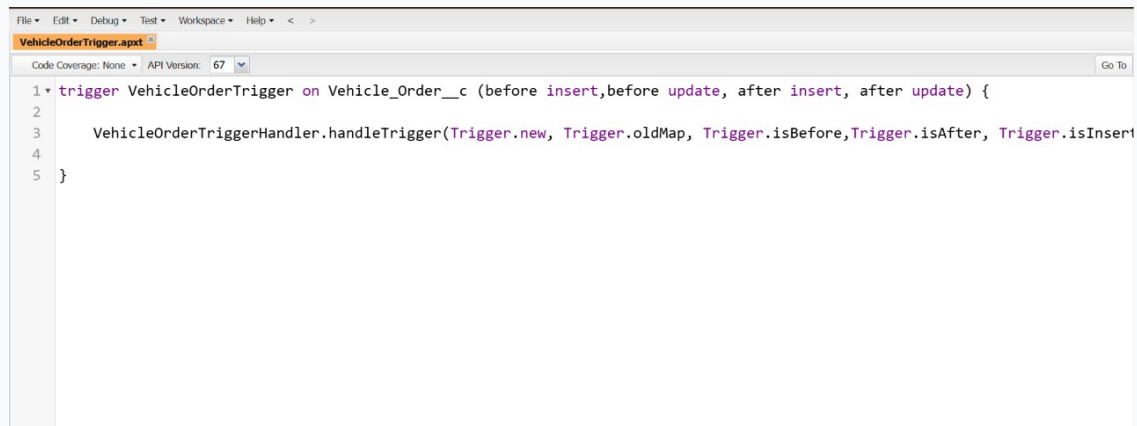
7.1 Apex Design

Apex is used for business rules that require transaction-level control. The project separates the trigger definition from the main processing logic by using a Trigger Handler class. This keeps the trigger small and makes the business logic easier to maintain.

Component	Responsibility
VehicleOrderTrigger	Invokes the handler for before/after insert and update events
VehicleOrderTriggerHandler	Contains stock validation and stock update logic
preventOrderIfOutOfStock()	Prevents orders when vehicle stock is unavailable
updateStockOnOrderPlacement()	Reduces stock for confirmed orders

7.2 Trigger Events

Event	Purpose
Before Insert	Validate stock before a new order is saved
Before Update	Validate stock when an order is changed
After Insert	Process stock update after order creation
After Update	Process stock update after order update



7.3 Trigger Handler Flow



7.4 Stock Validation

The handler collects the Vehicle IDs from the incoming orders, queries the related Vehicle records, and checks Stock__c. If the stock is not available, addError() is used to stop the order transaction.

7.5 Stock Reduction

For confirmed orders, the handler identifies the related vehicle and reduces Stock__c by one. The implementation collects records and performs the update in bulk.

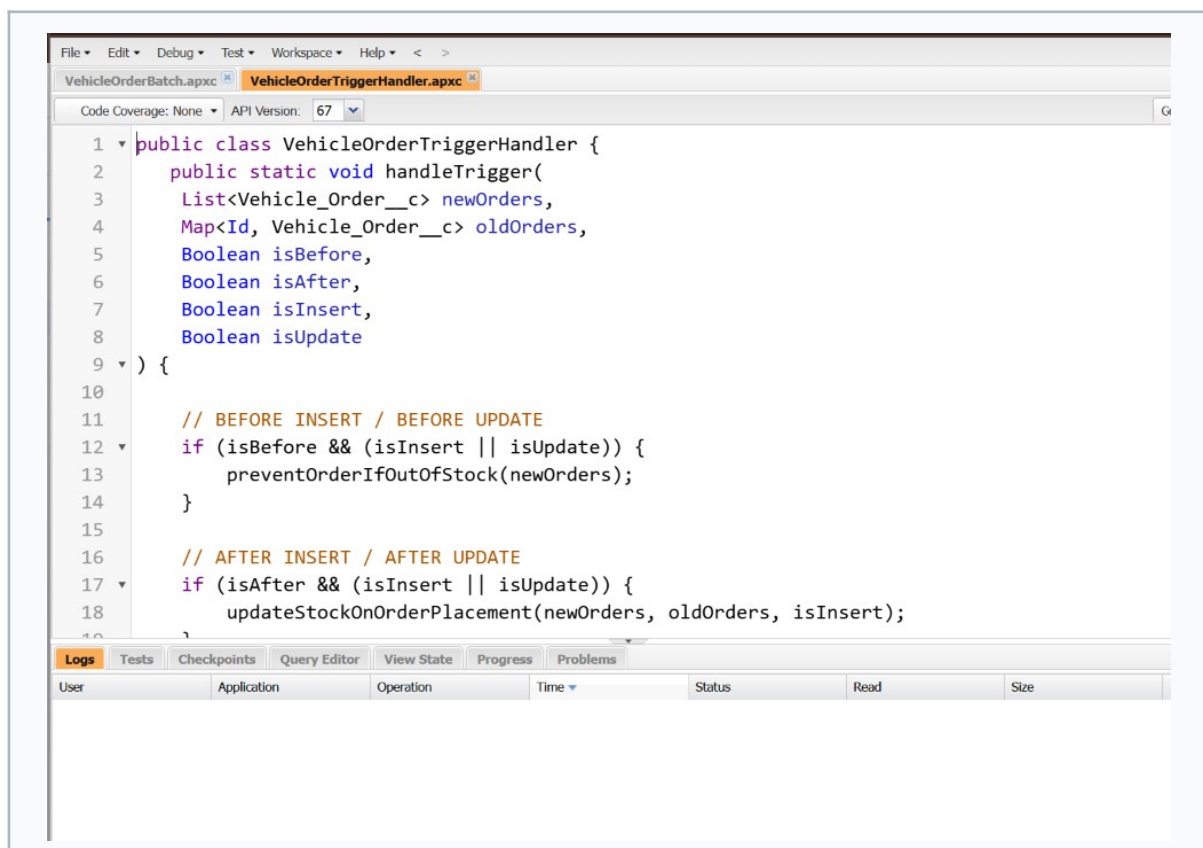


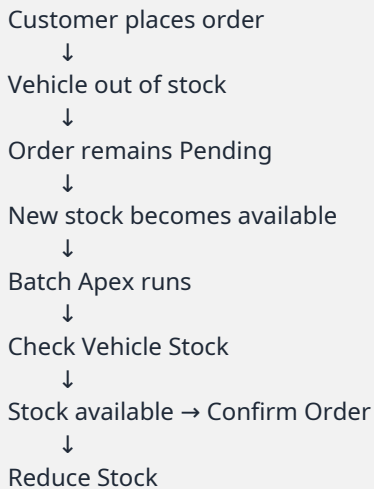
Figure 13 VehicleOrderTriggerHandler Apex Class

Figure 14 VehicleOrderTrigger Apex Trigger

8 Batch Apex and Scheduled Processing

8.1 Business Scenario

A customer may place an order when the selected vehicle has no available stock. In that situation, the order remains Pending rather than being lost. After new stock is added, the Batch Apex process checks pending orders and confirms eligible orders.

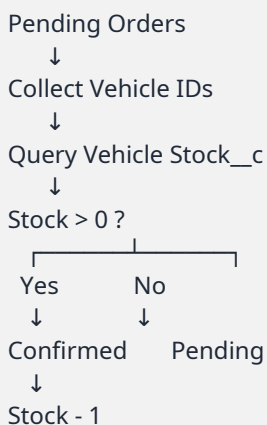


8.2 VehicleOrderBatch

The VehicleOrderBatch class implements the Salesforce Batchable interface and processes pending Vehicle Order records in batches. The start method selects pending orders, execute checks vehicle stock and updates eligible records, and finish records that the batch has completed.

Batch Method	Responsibility
start()	Selects Vehicle_Order__c records with Status__c = Pending
execute()	Checks vehicle stock, confirms eligible orders, and updates stock
finish()	Records completion of the batch job

8.3 Batch Processing Logic



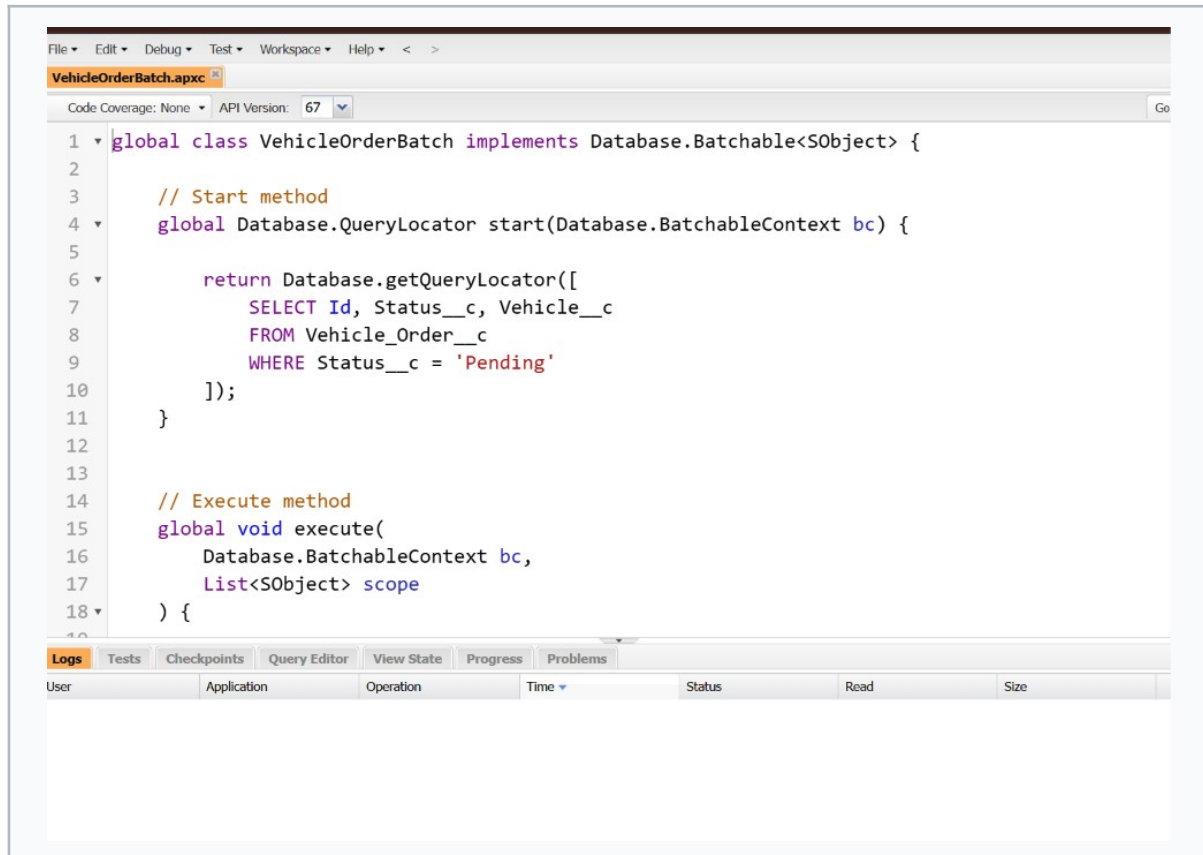


Figure 15 VehicleOrderBatch Apex Class

8.4 Scheduled Apex

A separate scheduler class, VehicleOrderBatchScheduler, is used to start the batch job automatically. The scheduler invokes Database.executeBatch() with a batch size of 50.

```

VehicleOrderBatchScheduler
↓
VehicleOrderBatch
↓
Process Pending Orders

```

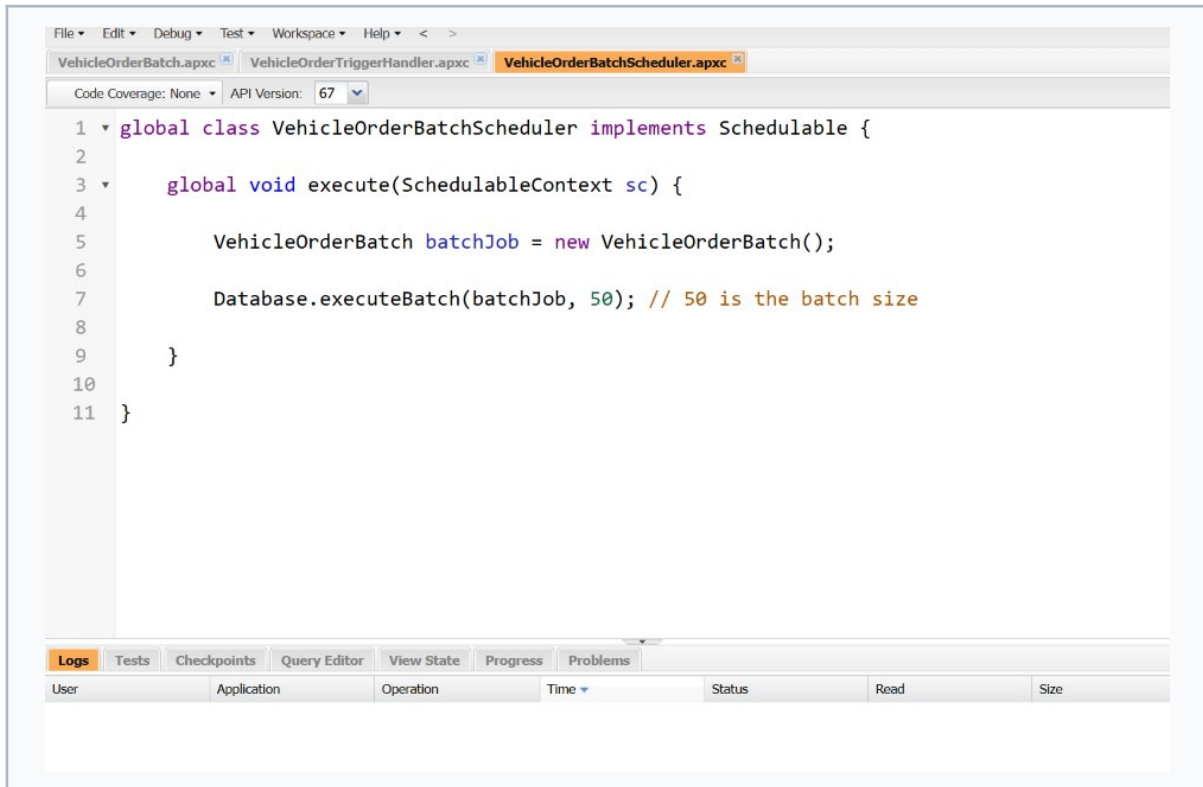


Figure 16 VehicleOrderBatchScheduler Apex Class

Figure 17 Scheduled Job Result

Evidence screenshot: Scheduled Jobs result to be inserted after the scheduled job is verified.

8.5 Scheduled Job

The batch can be scheduled to run automatically through Salesforce Scheduled Jobs. If the intended schedule is midnight, the Cron expression is 0 0 0 * * ?. The previously used expression 0 0 12 * * ? represents 12:00 PM.

9 User Interface and Sample Records

The WhatsNext Vision Motors Lightning App provides a central workspace for users to access the custom objects. Sample records can be used to demonstrate the relationships and automation implemented in the project.

Recommended Sample Records

Record Type	Useful Evidence
Vehicle	Vehicle name, model, Stock__c, price, dealer, status
Vehicle Dealer	Dealer name, location, code, phone, email
Vehicle Customer	Customer name, email, phone, address, preferred vehicle type
Vehicle Order	Customer, vehicle, order date, status, dealer
Vehicle Test Drive	Customer, vehicle, date, status

Record Type	Useful Evidence
Service Request	Customer, vehicle, service date, issue, status

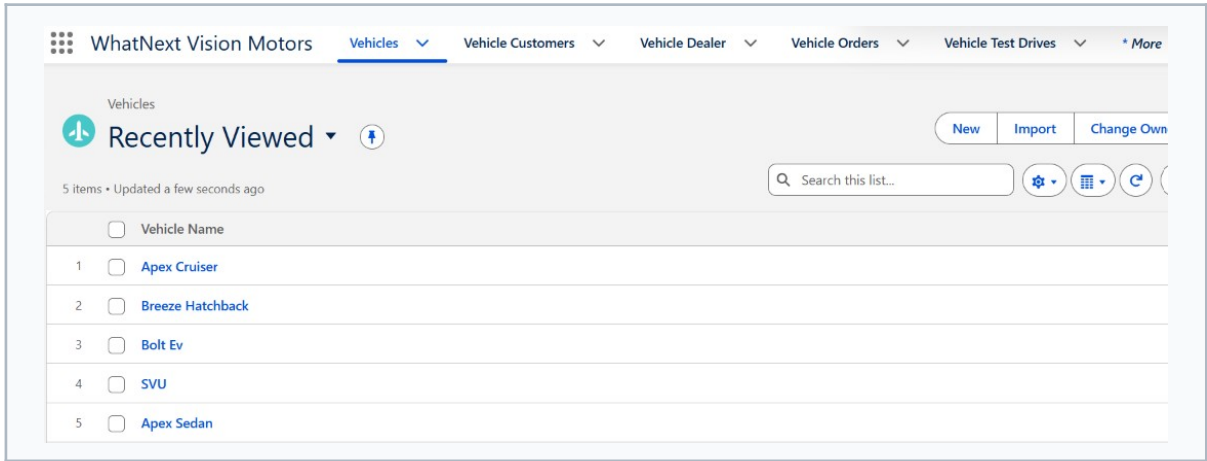


Figure 18 Vehicle Sample Record

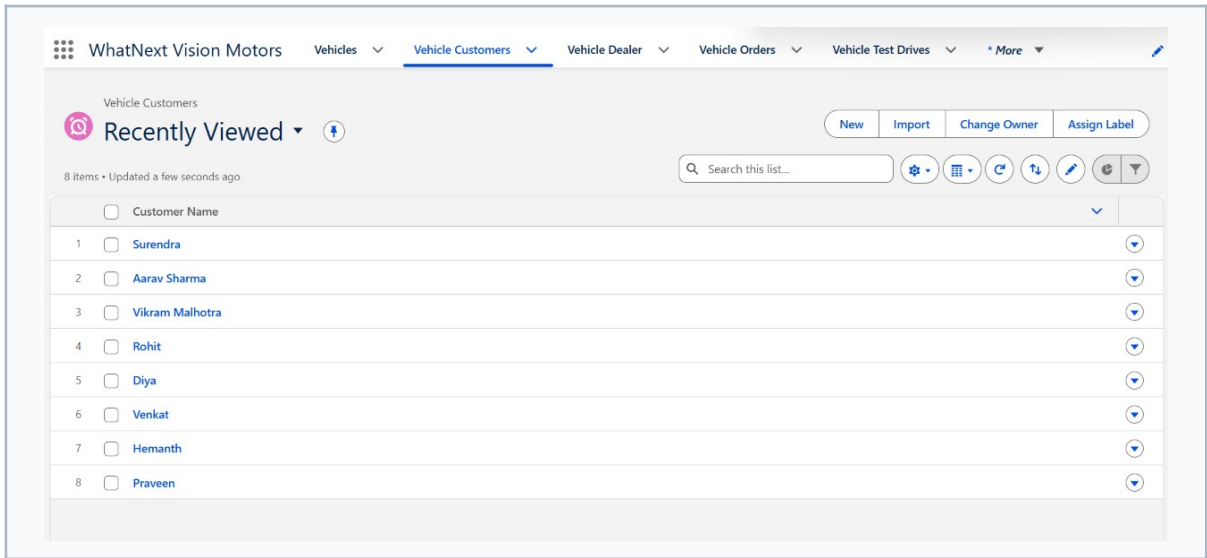


Figure 19 Vehicle Customer Sample Record

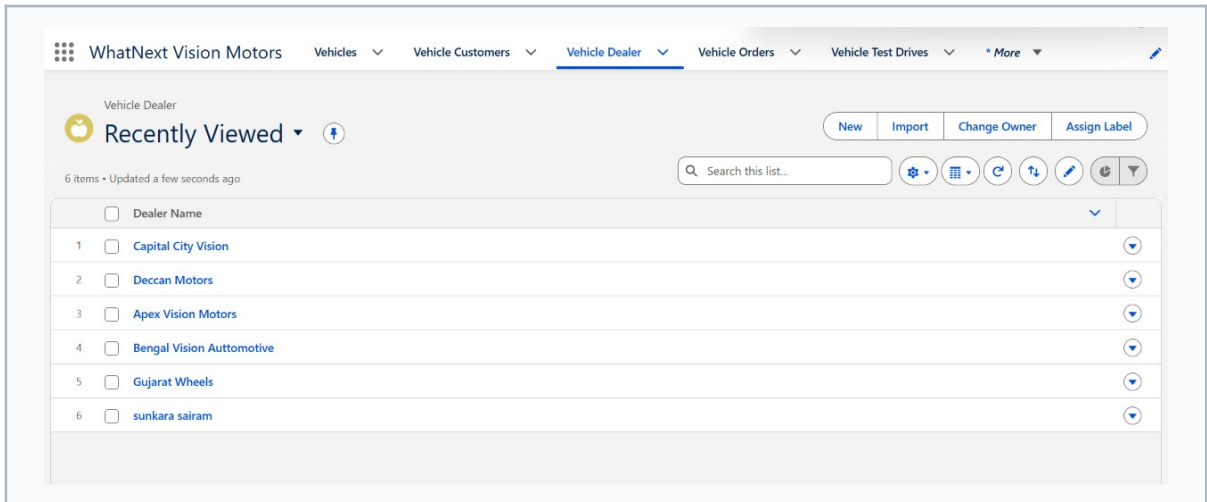


Figure 20 Vehicle Dealer Sample Record

WhatNext Vision Motors

VehiclesVehicle CustomersVehicle DealerVehicle OrdersVehicle Test DrivesMore

Vehicle Orders

Recently Viewed

NewImportChange OwnerAssign Label

9 items • Updated a few seconds ago

Search this list...

Vehicle Order No

1ORD-0009

2ORD-0008

3ORD-0007

4ORD-0006

5ORD-0005

6ORD-0001

7ORD-0004

8ORD-0003

9ORD-0002

Figure 21 Vehicle Order Sample Record

WhatNext Vision Motors

VehiclesVehicle CustomersVehicle DealerVehicle OrdersVehicle Test DrivesMore

Vehicle Test Drives

Recently Viewed

NewImportChange OwnerAssign Label

8 items • Updated a few seconds ago

Search this list...

Test Drive No

1TD-0008

2TD-0007

3TD-0006

4TD-0005

5TD-0004

6TD-0003

7TD-0002

8TD-0001

Figure 22 Vehicle Test Drive Sample Record

WhatNext Vision Motors

VehiclesVehicle CustomersVehicle DealerVehicle OrdersVehicle Service RequestMore

Vehicle Service Request

Recently Viewed

NewImportChange OwnerAssign Label

10 items • Updated a few seconds ago

Search this list...

Service Request No

1SR-0010

2SR-0009

3SR-0008

4SR-0007

5SR-0006

6SR-0005

7SR-0004

8SR-0003

9SR-0002

10SR-0001

Figure 23 Vehicle Service Request Sample Record

10 Functional Testing and Results

Functional testing is used to verify that the major business requirements behave as expected. The following test cases can be demonstrated using sample Salesforce records and the automation created for the project.

Test Case	Action	Expected Result	Evidence
Out-of-stock validation	Create an order for a vehicle with Stock__c = 0	Order is blocked with an error	Order error screenshot
Available vehicle order	Create an order for a vehicle with stock	Order can proceed	Order record
Stock deduction	Confirm an order	Vehicle stock decreases by one	Vehicle before/after
Dealer assignment	Create a Pending order	Relevant dealer is associated with the order	Order record
Test-drive reminder	Schedule a test drive	Reminder is prepared for one day before the date	Flow/email evidence
Pending order processing	Add stock for a pending order	Batch can confirm the eligible order	Order before/after
Scheduled processing	Run scheduled job	Batch job is invoked	Scheduled Jobs/Apex Jobs

10.1 Out-of-Stock Validation

A vehicle with Stock__c set to zero is used to test the stock validation rule. When the user attempts to create the order, the Trigger Handler calls addError() and prevents the transaction.

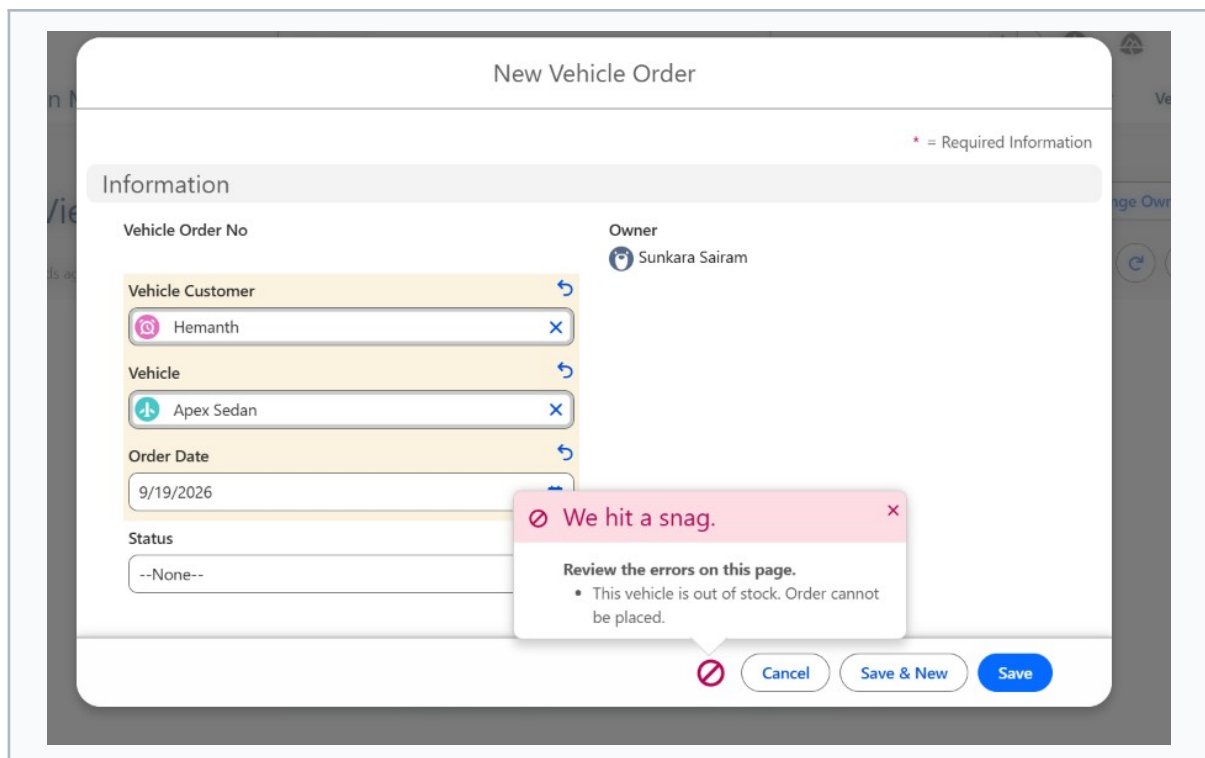


Figure 24 Out-of-Stock Validation Error

10.2 Stock Deduction Test

A confirmed order is used to verify that the related vehicle stock is reduced by one. The before-and-after records provide evidence that the Apex logic updated the inventory.

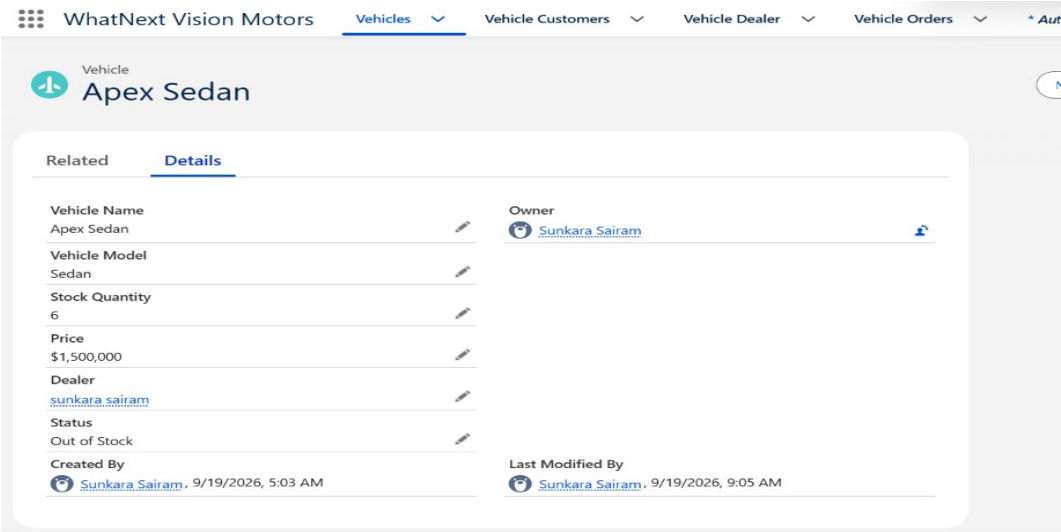
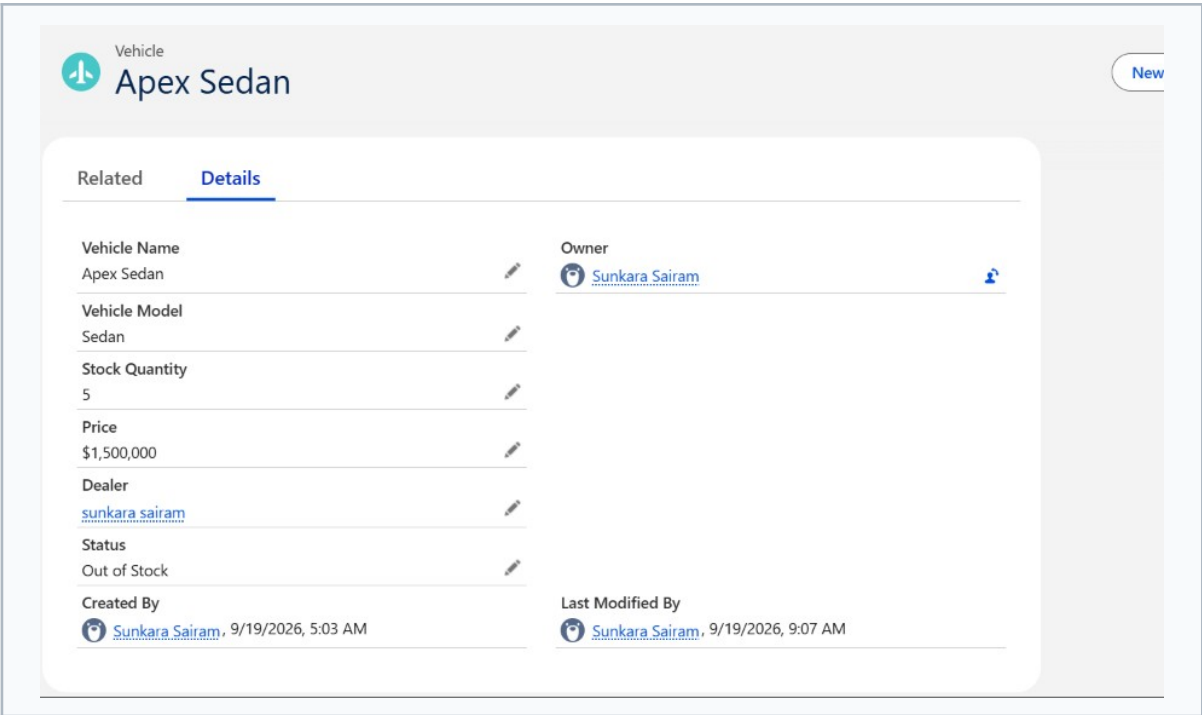


Figure 25 Vehicle Stock Before and After Confirmation



10.3 Dealer Assignment Test

A Pending order is created with a customer. The Flow retrieves the customer information, matches dealer location information, and associates the relevant dealer with the order.

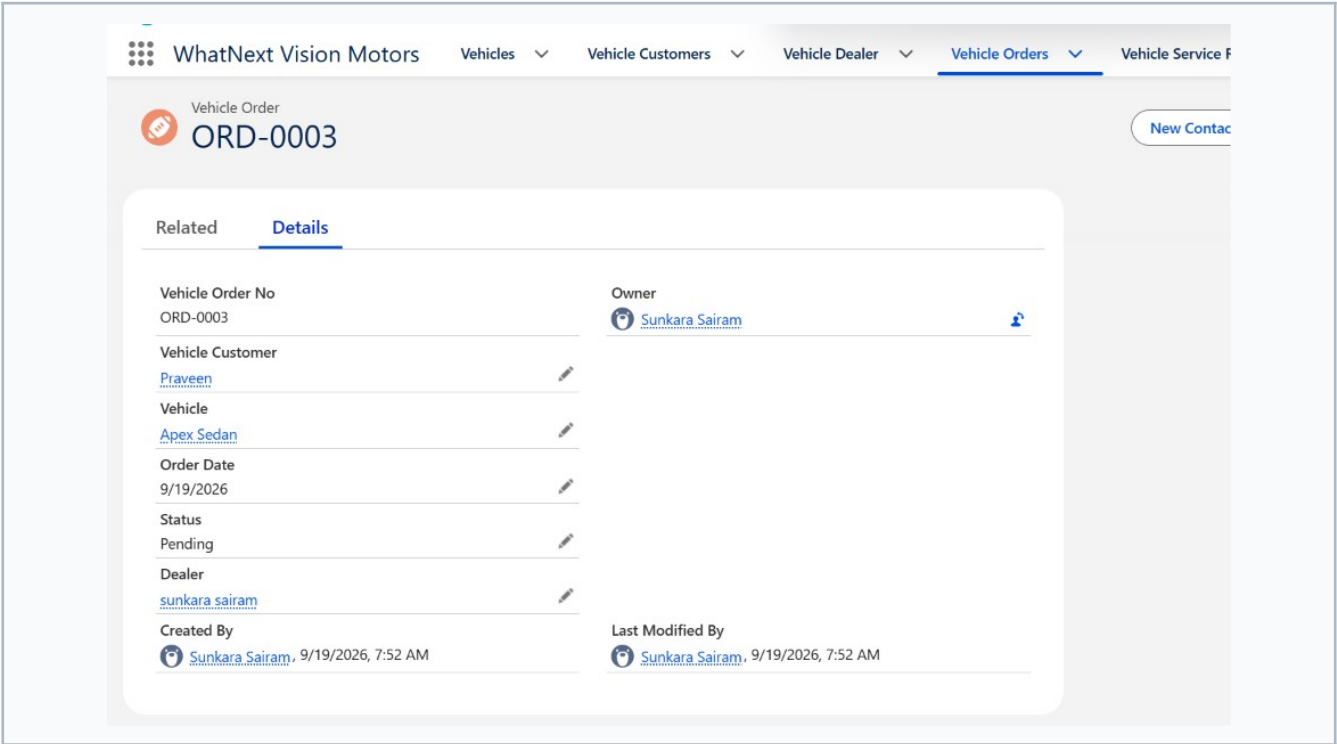


Figure 26 Dealer Assignment Test Result

10.4 Test Drive Reminder Test

A test drive is created with Scheduled status and a future test-drive date. The scheduled path is configured to run one day before the date and send the reminder email to the related customer.

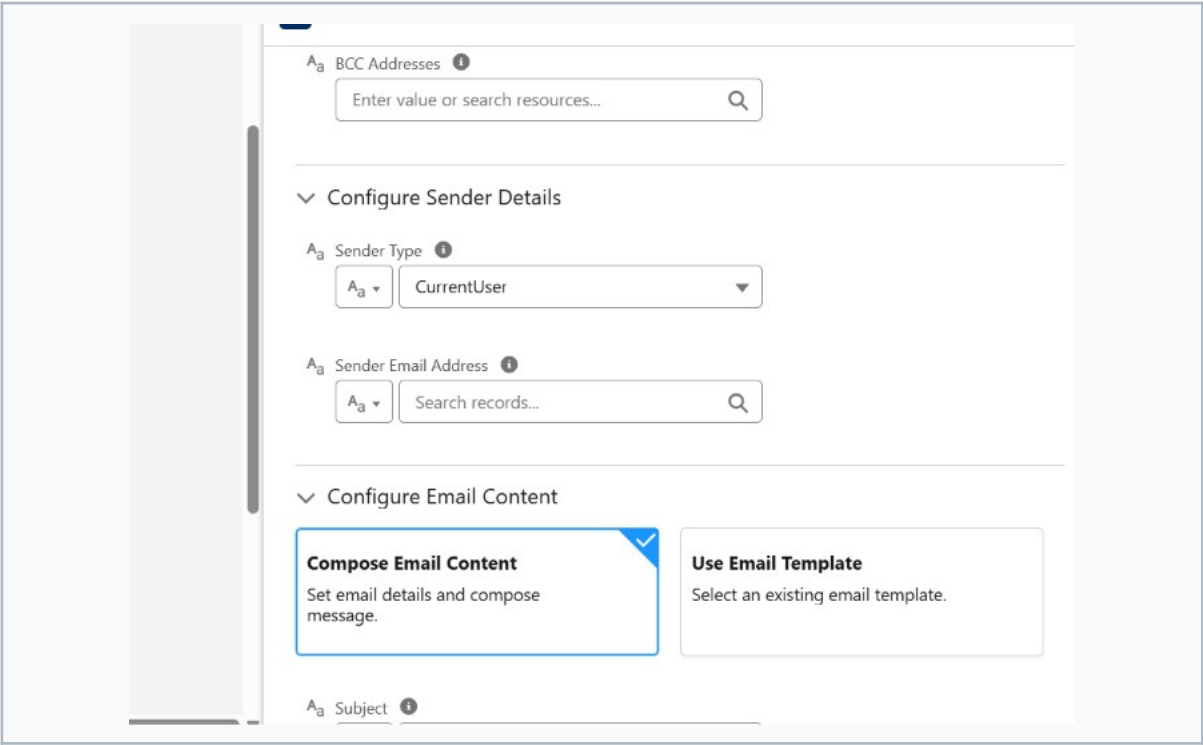


Figure 27 Test Drive Reminder Evidence

10.5 Batch Processing Test

A pending order is created while vehicle stock is unavailable. After stock is replenished, the batch job is executed. The expected result is that the eligible order changes to Confirmed and vehicle stock is reduced.

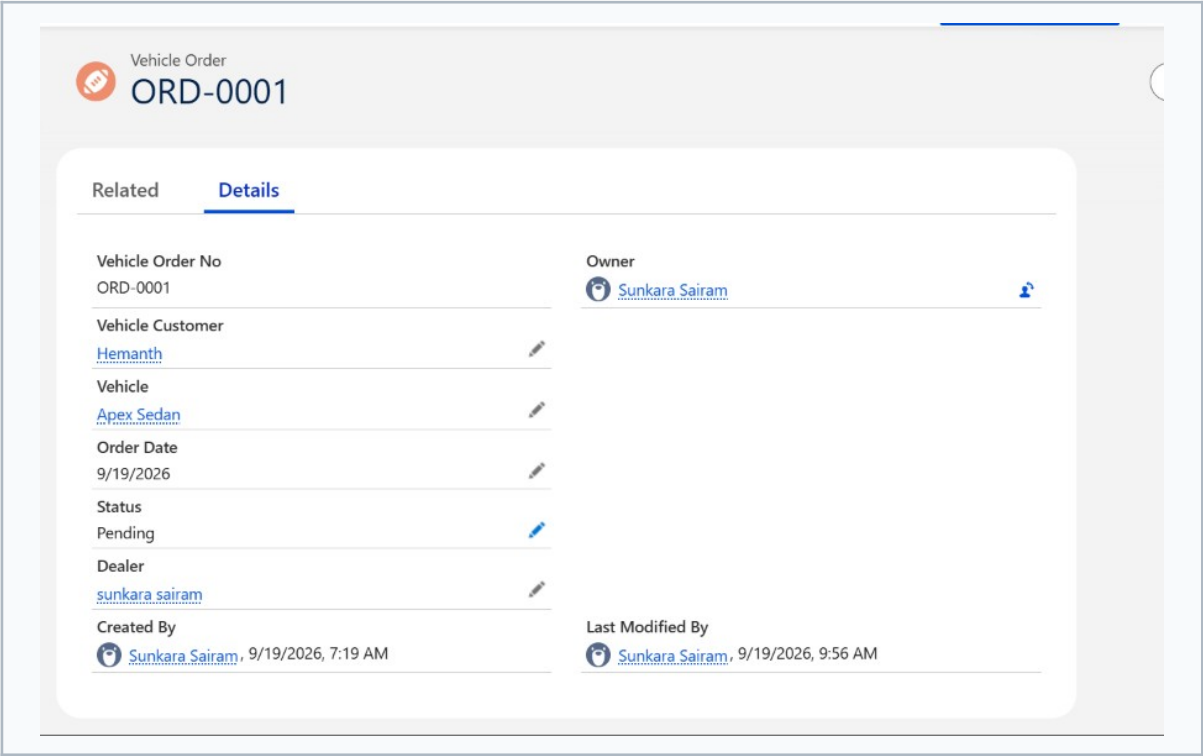


Figure 28 Pending Order Before Batch Processing

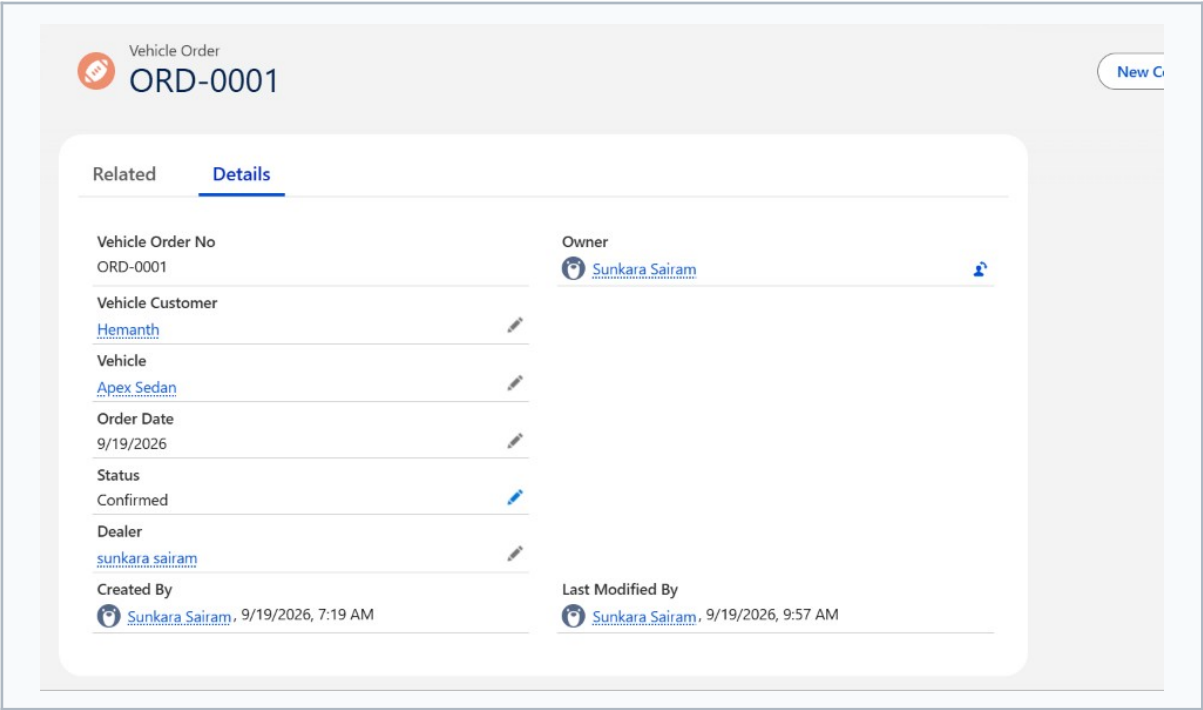


Figure 29 Order After Batch Processing

10.6 Apex and Scheduled Job Evidence

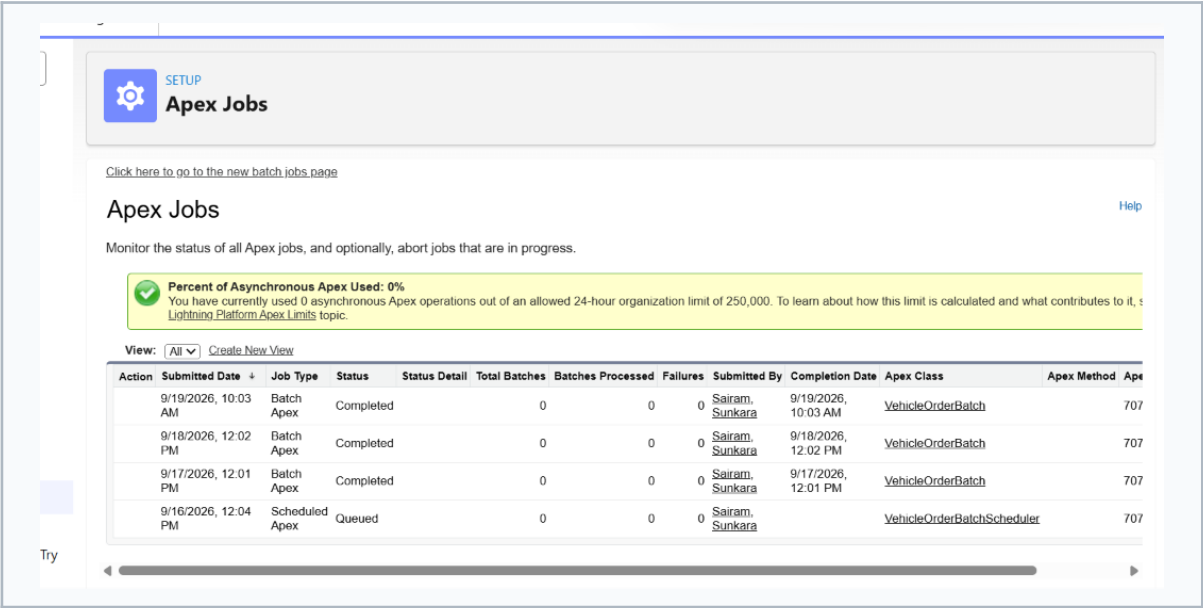


Figure 30 Apex Jobs Result

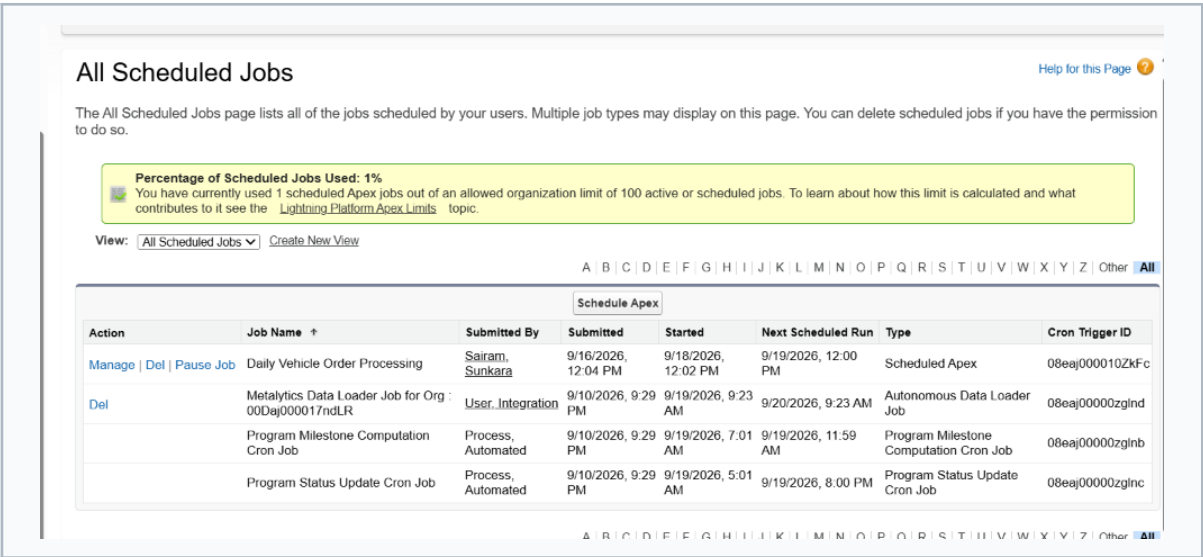


Figure 31 Scheduled Job Result

11 Conclusion

The WhatsNext Vision Motors Salesforce project provides a centralized solution for managing key automotive operations such as vehicle inventory, dealer information, customer records, vehicle orders, test drives, and service requests.

The project combines Salesforce configuration with programmatic development to automate important business processes. Custom objects and lookup relationships provide the foundation for the data model, while the Lightning App gives users a single workspace for accessing the records.

Record-Triggered Flows automate dealer assignment and test-drive reminders. Apex Trigger and Trigger Handler logic enforce the vehicle stock rule and update inventory when orders are confirmed. Batch Apex provides a bulk-processing mechanism for pending orders, and Scheduled Apex allows that processing to run automatically.

Overall, the project demonstrates practical experience with Salesforce custom objects, fields and relationships, custom tabs, Lightning App configuration, Record-Triggered Flows, Scheduled Paths, Apex Triggers, Trigger Handler design, Batch Apex, Scheduled Apex, and functional testing.

The implementation shows how Salesforce can be used as both a CRM platform and an application-development platform to automate real-world automotive business processes.